

3 (a) (i) speed = 4.0 m s^{-1} ...(*allow 1 s.f.*) A1 [1]

(ii) $v^2 = 2gh$

$= 2 \times 9.8 \times 1.96$ M1

$v = 6.2 \text{ m s}^{-1}$ A0 [1]

(*use of $g = 10 \text{ m s}^{-2}$ loses the mark*)

(b) correct basic shape with correct directions for vectors M1

speed = $(7.4 \pm 0.2) \text{ m s}^{-1}$ A1

at $(33 \pm 2)^\circ$ to the vertical A1 [3]

(*for credit to be awarded, speed and angle must be correct on the diagram – not calculated*)

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- (c) (i) either $v^2 = 2 \times 9.8 \times 0.98$ or $v = 6.2 / \sqrt{2}$
 speed = 4.4 m s^{-1}
 (allow calculation of $t = 0.447 \text{ s}$, then $v = 4.4 \text{ m s}^{-1}$)
- (ii) 1 momentum = mv C1
 change in momentum = $0.034 (6.2 + 4.4)$ C1
 = 0.36 kg m s^{-1} A1 [3]
 (use of $0.034 (6.2 - 4.4)$ loses last two marks)
- 2 force = $\Delta p / \Delta t$ (however expressed) C1
 = $\frac{0.36}{0.12}$
 = 3.0 N (allow 1 s.f.) A1 [2]

[Total: 12]